DSA Sheet by AlgoForge

Contact us here: algoforge.co
https://t.me/samratIsLive

Always remember why you started. All the very best

Make a copy of this sheet

Introduction to any Coding language

- [ ] Data types (Primitive and non-primitve) — https://www.w3schools.com/java/java_data_types.asp
- [ ] How to take input output — https://www.w3schools.com/java/java_user_input.asp
- [ ] Find average of 3 numbers — https://beginnersbook.com/2020/01/java-program-to-find-average-of-3-numbers/

Conditionals

- [ ] Conditionals, if-else statement — Java Conditions and If Statements
- [ ] Check postive or negative number — Java: Get a number from the user and print whether it is positive or negative number
- [ ] Day of week — https://www.w3resource.com/java-exercises/conditional-statement/java-conditional-statement-exercise-5.php
- [ ] Find if year is leap or not — Java Program to Find if a Given Year is a Leap Year - GeeksforGeeks

- [ ] Operators — https://www.w3schools.com/java/java_operators.asp

Loops

- [ ] While loop, for loop, do while loop — https://www.geeksforgeeks.org/java/loops-in-java/
- [ ] Check if a stream of number is even or odd — https://www.geeksforgeeks.org/problems/odd-or-even3618/1
- [ ] Find factorial of a number — https://www.geeksforgeeks.org/problems/factorial5739/1
- [ ] Check if a number is prime or not in N iterations — https://www.geeksforgeeks.org/java/java-prime-number-program/
- [ ] Check if a number is prime or not in root(N) iterations — https://www.geeksforgeeks.org/java/java-prime-number-program/
- [ ] Print all prime numbers less than N — https://www.geeksforgeeks.org/dsa/print-all-prime-numbers-less-than-or-equal-to-n/

Number System

- [ ] Print fibonacci series — https://www.geeksforgeeks.org/java/java-fibonacci-series/
- [ ] Count digits in a number — https://www.geeksforgeeks.org/problems/count-digits-1606889545/1
- [ ] Find sum of digits of a number — https://www.geeksforgeeks.org/problems/sum-of-digits1742/1
- [ ] Reverse a number — https://www.geeksforgeeks.org/java/java-reverse-number-program/
- [ ] Rotate a number by k — https://www.geeksforgeeks.org/dsa/rotate-digits-of-a-given-number-by-k/
- [ ] Find GCD of two numbers — https://www.geeksforgeeks.org/problems/gcd-of-two-numbers3459/1
- [ ] Find LCM of two numbers — https://www.geeksforgeeks.org/problems/lcm-of-two-numbers/1

Patterns

- [ ] Print right angle triangle — https://www.geeksforgeeks.org/java/java-program-to-print-right-triangle-star-pattern/
- [ ] Downward triangle star pattern — https://www.geeksforgeeks.org/java/java-program-to-print-downward-triangle-star-pattern/
- [ ] Left right angle triangle pattern — https://www.geeksforgeeks.org/java/java-program-to-print-left-triangle-star-pattern/
- [ ] Diamond star pattern — https://www.geeksforgeeks.org/java/java-program-to-print-diamond-shape-star-pattern/
- [ ] Print star diagonals pattern — https://www.geeksforgeeks.org/dsa/program-print-diagonal-star-patterns/
- [ ] Hollow diamond pattern — https://www.geeksforgeeks.org/dsa/program-print-hollow-pyramid-diamond-pattern/#:~:text=2-,.%20Hollow%20Diamond,-Note%3A%20For
- [ ] Number changing pyramid pattern — https://www.geeksforgeeks.org/java/java-pattern-programs/#5-numberchanging-pyramid-pattern:~:text=1%202%20%0A1-,5.,-Number%2DChanging%20Pyramid
- [ ] Fibonacci pyramid pattern — https://www.geeksforgeeks.org/dsa/program-print-fibonacci-triangle/
- [ ] Diamond number pattern — https://www.geeksforgeeks.org/dsa/java-program-to-print-numbers-with-diamond-pattern/
- [ ] Butterfly half pattern — https://www.geeksforgeeks.org/dsa/program-to-print-butterfly-pattern-star-pattern/
- [ ] Hollow hourglass — https://www.geeksforgeeks.org/java/java-pattern-programs/#22-hollow-hourglass-pattern:~:text=*%0A%20%20%20%20*%20*%0A%20%20%20*%20%20%20*%0A%20*[illegible]*%0A%20%20%20*%20%20%20*%0A%20%20*%20%20%20%20%20*%0A%20*%20%20%20%20%20%20%20*%0A*%20%20%20%20%20%20%20%20%20*%0A%20*%20%20%20%20%20%20%20*%0A[illegible]%20%20%20%20%20%20*%0A%20%20*%20%20%20%20%20*%0A%20%20%20*%20%20%20*%0A%20%20%20%20*%20*%0A%20%20%20%20%20*-,22.-,%20Hollow%20Hourglass%20P

Functions and Arrays

- [ ] What are functions — https://www.youtube.com/watch?v=HG8tV0zCsBQ
- [ ] Frequency of a digit — https://www.geeksforgeeks.org/dsa/find-the-frequency-of-a-digit-in-a-number/
- [ ] Convert a decimal base to any base (Use integers as inputs) — https://www.geeksforgeeks.org/dsa/convert-base-decimal-vice-versa/
- [ ] Convert any base to decimal base (Use integers as inputs) — https://www.geeksforgeeks.org/dsa/convert-base-decimal-vice-versa/
- [ ] Calculate NcR — https://www.geeksforgeeks.org/problems/ncr1019/1

Arrays

Just the introduction, more questions are of arrays and strings at the end

- [ ] Arrays memory allocation in Java — https://medium.com/@f2015939p/array-memory-allocation-in-java-49b292122d18
- [ ] Find number of elements greater than x
- [ ] Search in a unsorted array — https://www.geeksforgeeks.org/problems/search-an-element-in-an-array-1587115621/1
- [ ] Find the difference between maximum and minimum element of array — https://www.geeksforgeeks.org/problems/find-minimum-and-maximum-element-in-an-array4428/1
- [ ] Add two arrays — https://www.geeksforgeeks.org/problems/add-two-numbers-represented-by-two-arrays2408/1
- [ ] Subtract two arrays
- [ ] Reverse an array — https://www.geeksforgeeks.org/java/reverse-an-array-in-java/
- [ ] Rotate an array — https://leetcode.com/problems/rotate-array/description/
- [ ] Print all subarrays
- [ ] Count subarrays having sum K (Use brute force)
- [ ] Count subarrays having sum divisible by K (Use brute force)
- [ ] Find max sum subarray (Kadane's algorithm)
- [ ] Print all subsets of array

- [ ] Binary search implementation — https://www.geeksforgeeks.org/dsa/binary-search/
- [ ] Find floor value of a given target — https://www.geeksforgeeks.org/problems/floor-in-a-sorted-array-1587115620/1
- [ ] Find ceil value of a given target — https://www.geeksforgeeks.org/problems/ceil-in-a-sorted-array/1
- [ ] Find first and last position of element — Find First and Last Position of Element in Sorted Array - LeetCode

2D Arrays

- [ ] Introduction to 2D arrays — https://www.geeksforgeeks.org/dsa/introduction-to-matrix-or-grid-data-structure-and-algorithms-tutorial/
- [ ] Multiply 2 matrices — https://www.geeksforgeeks.org/problems/multiply-2-matrices4144/1
- [ ] Wave traversal
- [ ] Exit point of matrix — https://www.geeksforgeeks.org/problems/exit-point-in-a-matrix0905/1
- [ ] Spiral matrix — https://leetcode.com/problems/spiral-matrix/description/
- [ ] Rotate matrix — https://leetcode.com/problems/rotate-image/description/
- [ ] Search a 2d matrix — https://leetcode.com/problems/search-a-2d-matrix/description/
- [ ] Search a 2d matrix 2 — https://leetcode.com/problems/search-a-2d-matrix-ii/description/
- [ ] Diagonal traversal of matrix

Arraylist and Strings

Just introduction, more questions at end

- [ ] Arraylist Introduction — https://www.w3schools.com/java/java_arraylist.asp
- [ ] String interning — https://www.geeksforgeeks.org/java/interning-of-string/
- [ ] Compress String by omitting consecutive repeating characters
- [ ] Compress string by concatenationg consecutive character count
- [ ] String compression — https://leetcode.com/problems/string-compression/description/
- [ ] Toggle casing of characters — https://www.geeksforgeeks.org/dsa/program-toggle-characters-string/

Recursion and Backtracking

- [ ] Understanding call stack — https://dev.to/theplebdev/java-quickies-the-call-stack-3g4g
- [ ] Swap two numbers using function
- [ ] Print decreasing numbers from n to 1
- [ ] Print increasing numbers from 1 to n
- [ ] Print decreasing from n to 1, then 1 to n in one recursive function
- [ ] Finf factorial of a number — https://www.geeksforgeeks.org/problems/factorial5739/1
- [ ] Find power of a number — https://www.geeksforgeeks.org/problems/power-of-numbers-1587115620/1
- [ ] Find power logarithmic — https://www.geeksforgeeks.org/problems/power-of-numbers-1587115620/1
- [ ] Understand pre, in and post area of recursive calls
- [ ] Tower of hanoi — https://www.geeksforgeeks.org/problems/tower-of-hanoi-1587115621/1

Recursion on Arrays

- [ ] Print array using recursion
- [ ] Print array in reverse
- [ ] Find max element in array using recursion
- [ ] Find first index of a target element
- [ ] Find last index of a target element, initial input in function will be idx = 0
- [ ] Return an array consisting of all indices of target element

Recursion on the way Down

- [ ] Return arraylist of all the subsequence of a string — https://www.geeksforgeeks.org/dsa/print-all-subsequences-of-a-string-using-arraylist/
- [ ] Get all keypad combinations — https://www.geeksforgeeks.org/problems/possible-words-from-phone-digits-1587115620/1
- [ ] Get all stair paths, 1-2-3 steps allowed at a time
- [ ] Get all maze paths if only right and down step allowed
- [ ] Get all maze paths if only right and down step with jumps allowed

Recursion on the way Up

- [ ] Print all subsequences of a string — https://www.geeksforgeeks.org/dsa/print-subsequences-string/
- [ ] Print all keypad combinations — https://www.geeksforgeeks.org/problems/possible-words-from-phone-digits-1587115620/1
- [ ] Print all stair paths, 1-2-3 steps allowed at a time
- [ ] Print all maze paths if only right and down step allowed
- [ ] Print all maze paths if only right and down step with jumps allowed

- [ ] Print Encodings of a string
- [ ] Print permutations of a string
- [ ] Get Max path sum in a maze — https://www.geeksforgeeks.org/problems/path-in-matrix3805/1
- [ ] Count friends pairing — https://www.geeksforgeeks.org/problems/friends-pairing-problem5425/1
- [ ] Josephus Problem — https://www.geeksforgeeks.org/problems/josephus-problem/1

Backtracking

- [ ] Flood fill backtracking — (Go from 0,0 to n-1,m-1), you can travel on 1, not on you and can move in 4 directions
- [ ] Coin change permutation (Generate all the possible answers) — https://www.geeksforgeeks.org/problems/number-of-coins1824/1 (this question but one coin can be used one time)
- [ ] Coin change permutation mutliple use (Generate all the possible answers) — https://www.geeksforgeeks.org/problems/number-of-coins1824/1
- [ ] Coin change combination multiple use (Generate all the possible answers) — https://leetcode.com/problems/coin-change-ii/description/ (this question but use one coin one time)
- [ ] Coin change combination multiple use (Generate all the possible answers) — https://leetcode.com/problems/coin-change-ii/description/
- [ ] Target sum subset (Generate all the possible subsets) — https://www.geeksforgeeks.org/problems/subset-sum-problem-1611555638/1
- [ ] Partition array into 2 equal subsets — https://leetcode.com/problems/partition-equal-subset-sum/description/
- [ ] N- Queens — https://leetcode.com/problems/n-queens/description/
- [ ] Check knight tour — https://leetcode.com/problems/check-knight-tour-configuration/description/
- [ ] Print all the knight tour possible answers
- [ ] Check if sudoku is valid — https://leetcode.com/problems/valid-sudoku/description/
- [ ] Sudoku Solver — https://leetcode.com/problems/sudoku-solver/description/
- [ ] N-queens optimised with O(N) space — https://leetcode.com/problems/n-queens/description/
- [ ] N-queens optimised with O(1) extra space — https://leetcode.com/problems/n-queens/description/
- [ ] Beautiful Arrangement — https://leetcode.com/problems/beautiful-arrangement/description/
- [ ] Word search — https://leetcode.com/problems/word-search/description/
- [ ] Verbal Arithmetic Puzzle (should have some hasmap knowledge) — https://leetcode.com/problems/verbal-arithmetic-puzzle/description/
- [ ] Word break 2 — https://leetcode.com/problems/word-break-ii/description/
- [ ] Word search 2 (skip for now if you don't know TRIE) — https://leetcode.com/problems/word-search-ii/description/

Object Oriented Programming

- [ ] Only one link is enough, go through it all, solve problems, read concepts — **https://www.tpointtech.com/java-oops-concepts**

Linked List

- [ ] What is Node for linkedlist — https://www.geeksforgeeks.org/java/implementing-a-linked-list-in-java-using-class/

Construction of our own LinkedList

Take help from here to understand more: https://github.com/Samrat-Bhardwaj/ALGOFORGE/blob/main/Batch1/LinkedList/Construction/Main.java

- [ ] Add a value at last in LinkedList
- [ ] Add a value at start in LinkedList
- [ ] Print the complete linkedlist
- [ ] Remove last node from linkedlist
- [ ] Remove first node from linkedlist
- [ ] Get Node at a particular index in LinkedList
- [ ] Add node at particular index in LinkedList
- [ ] Remove node from particular index
- [ ] Reverse Linked list by swapping data — https://leetcode.com/problems/reverse-linked-list/description/
- [ ] Reverse linkedlist by reversing pointers — https://leetcode.com/problems/reverse-linked-list/description/

Questions

- [ ] Get Kth node from end of LL — https://www.geeksforgeeks.org/problems/nth-node-from-end-of-linked-list/0
- [ ] Find middle of linkedlist using hare tortoise algo — https://leetcode.com/problems/middle-of-the-linked-list/description/
- [ ] Pallindrome Linkedlist — https://leetcode.com/problems/palindrome-linked-list/description/
- [ ] Reorder list — https://leetcode.com/problems/reorder-list/description/
- [ ] Undo re-ordering of list (Undo above question)
- [ ] Odd even linkedlist — https://leetcode.com/problems/odd-even-linked-list/description/
- [ ] Seggregate even odd lists — https://www.geeksforgeeks.org/problems/segregate-even-and-odd-nodes-in-a-linked-list5035/1
- [ ] Reverse nodes in k group — https://leetcode.com/problems/reverse-nodes-in-k-group/description/
- [ ] Copy List with random pointer — https://leetcode.com/problems/copy-list-with-random-pointer/description/
- [ ] Add two LL — https://leetcode.com/problems/add-two-numbers/description/
- [ ] Remove duplicates from sorted LL, keeping first duplicate — https://leetcode.com/problems/remove-duplicates-from-sorted-list/description/
- [ ] Remove all duplicates from LL — https://leetcode.com/problems/remove-duplicates-from-sorted-list-ii/description/

- [ ] Linkedlist cycle — https://leetcode.com/problems/linked-list-cycle/description/
- [ ] Starting point of cycle in linkedList — https://www.geeksforgeeks.org/problems/find-the-first-node-of-loop-in-linked-list--170645/1
- [ ] Merge two sorted lists — https://leetcode.com/problems/merge-two-sorted-lists/description/
- [ ] Intersection of two LL — https://leetcode.com/problems/intersection-of-two-linked-lists/description/

- [ ] Design browser history — https://leetcode.com/problems/design-browser-history/description/
- [ ] Reverse nodes in even length group — https://leetcode.com/problems/reverse-nodes-in-even-length-groups/description/
- [ ] Rotate lists — https://leetcode.com/problems/rotate-list/description/
- [ ] Merge in between linkedlist — https://leetcode.com/problems/merge-in-between-linked-lists/
- [ ] Introduction of circular LL is enough after all these problems — https://www.geeksforgeeks.org/dsa/circular-linked-list/

Stack

- [ ] Duplicate parentheses — https://www.geeksforgeeks.org/find-expression-duplicate-parenthesis-not/
- [ ] Valid parentheses — https://leetcode.com/problems/valid-parentheses/
- [ ] Next greater element on right method 1 — https://www.geeksforgeeks.org/problems/next-larger-element-1587115620/1
- [ ] Next greater on right from Left to right — https://www.geeksforgeeks.org/problems/next-larger-element-1587115620/1
- [ ] Next Smaller on left — https://www.geeksforgeeks.org/problems/smallest-number-on-left3403/1
- [ ] Next smaller on left, iterate from right — https://www.geeksforgeeks.org/problems/smallest-number-on-left3403/1
- [ ] Next smaller on right, iterate from left — https://www.geeksforgeeks.org/next-smaller-element/
- [ ] Next smaller on right, iterate from right — https://www.geeksforgeeks.org/next-smaller-element/
- [ ] Stock Span — https://www.geeksforgeeks.org/problems/stock-span-problem-1587115621/1
- [ ] Stock span in class — https://leetcode.com/problems/online-stock-span/description/
- [ ] Largest area histogram — https://leetcode.com/problems/largest-rectangle-in-histogram/description/
- [ ] Largest area histogram opitimized — https://leetcode.com/problems/largest-rectangle-in-histogram/description/
- [ ] Maximal rectangle — https://leetcode.com/problems/maximal-rectangle/description/
- [ ] Sliding Window Maximum using stack — https://leetcode.com/problems/sliding-window-maximum/
- [ ] Inifix Evaluation — https://www.geeksforgeeks.org/problems/fun-with-expresions2523/1
- [ ] Basic calculator 2 — https://leetcode.com/problems/basic-calculator-ii/submissions/1629833449/
- [ ] Infix to prefix conversion — https://www.geeksforgeeks.org/convert-infix-prefix-notation/
- [ ] Infix to postfix conversion — https://www.geeksforgeeks.org/problems/infix-to-postfix-1587115620/1
- [ ] Prefix evaluation and conversion — https://www.geeksforgeeks.org/evaluation-prefix-expressions/
- [ ] Postfix evaluation and conversion — https://www.geeksforgeeks.org/evaluation-of-postfix-expression/?ref=rp, https://leetcode.com/problems/evaluate-reverse-polish-notation/description
- [ ] Merge Intervals — https://leetcode.com/problems/merge-intervals/description/
- [ ] Merge Intervals without stack — https://leetcode.com/problems/merge-intervals/description/
- [ ] Build static stack — https://www.geeksforgeeks.org/problems/implement-stack-using-array/1
- [ ] Build dynamic stack
- [ ] Build Static Queue
- [ ] Build Dynamic Queue
- [ ] Build stack using Queue push efficient — https://leetcode.com/problems/implement-stack-using-queues/
- [ ] Build stack using Queue pop efficient — https://leetcode.com/problems/implement-stack-using-queues/
- [ ] Build Queue using Stack add efficient — https://leetcode.com/problems/implement-queue-using-stacks/description/
- [ ] Build Queue using Stack remove efficient — https://leetcode.com/problems/implement-queue-using-stacks/description/
- [ ] Two stack in array — https://www.geeksforgeeks.org/problems/implement-two-stacks-in-an-array/1

Min Stack	https://leetcode.com/problems/min-stack/
Min Stack constant space	https://leetcode.com/problems/min-stack/
Next greater right in circular array	https://leetcode.com/problems/next-greater-element-ii/
Next greater right in circular array, move from right to left	https://leetcode.com/problems/next-greater-element-ii/
Validate stack sequences	https://leetcode.com/problems/validate-stack-sequences/description/
Validate stack sequences in O(1) Space	https://leetcode.com/problems/validate-stack-sequences/description/
Minimum add to make parentheses valid	https://leetcode.com/problems/minimum-add-to-make-parentheses-valid/description/
Remove outermost parentheses	https://leetcode.com/problems/remove-outermost-parentheses/description/
Minimum insertions to make parentheses valid 2	https://leetcode.com/problems/add-minimum-number-of-insertions-to-balance-a-parentheses-string
Valid Parentheses with *	https://leetcode.com/problems/valid-parenthesis-string/description/
Make Parentheses valid with locked array	https://leetcode.com/problems/check-if-a-parentheses-string-can-be-valid/description/
Parentheses Score	https://leetcode.com/problems/score-of-parentheses/
132 Pattern	https://leetcode.com/problems/132-pattern/
Asteroid Collision	https://leetcode.com/problems/asteroid-collision/description/
Remove K digits to make smallest number	https://leetcode.com/problems/remove-k-digits/description/
Simplify Path	https://leetcode.com/problems/simplify-path/description/?envType=problem-list-v2&envId=stack
Trapping rain water using stack	https://leetcode.com/problems/trapping-rain-water/description/
Trapping rain water using array	https://leetcode.com/problems/trapping-rain-water/description/
Trapping rain water O(1) Space	https://leetcode.com/problems/trapping-rain-water/description/
Container with most water using stack	https://leetcode.com/problems/container-with-most-water/description/
Container with most water in O(1) space	https://leetcode.com/problems/container-with-most-water/description/
Lexicographically smallest subset	https://leetcode.com/problems/find-the-most-competitive-subsequence/description/
Number of Valid Subarrays	https://leetcode.ca/all/1063.html
Basic Calculator 1	Leetcode question
Basic Calculator 3	Leetcode question
Basic Calculator 4	Leetcode question
Check if word is valid after substitutions	https://leetcode.com/problems/check-if-word-is-valid-after-substitutions/description/
Max Stack	https://www.lintcode.com/problem/656/

Queue

	Link
Number of island cells	https://leetcode.com/problems/number-of-islands/description/
Moving Average from Data Stream	https://www.lintcode.com/problem/642/
Hit counter	https://www.lintcode.com/problem/362/
Sliding Window Maximum using deque	https://leetcode.com/problems/sliding-window-maximum/
Shortest subarray with sum of least k	https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/
Design circular deque	https://leetcode.com/problems/design-circular-queue/description/
Design circular deQueue	https://leetcode.com/problems/design-circular-deque/

HashMap

	Link
Maximum frequency character	https://www.geeksforgeeks.org/problems/maximum-occuring-character-1587115620/1
Count elements with maximum frequency	https://leetcode.com/problems/count-elements-with-maximum-frequency/description/
Intersection of two arrays using map	https://leetcode.com/problems/intersection-of-two-arrays/
Intersection of two arrays using set	https://leetcode.com/problems/intersection-of-two-arrays/
All common elements of two arrays	https://www.geeksforgeeks.org/problems/common-elements5420/1
Common elements	https://leetcode.com/problems/find-common-elements-between-two-arrays/
Check if Array Pairs Are Divisible by k	https://leetcode.com/problems/check-if-array-pairs-are-divisible-by-k/description/
Count distinct elements in every window	https://www.geeksforgeeks.org/problems/count-distinct-elements-in-every-window/1
Largest subarray with zero sum	https://www.geeksforgeeks.org/problems/largest-subarray-with-0-sum/1
Count of subarray with zero sum	https://www.geeksforgeeks.org/problems/zero-sum-subarrays1825/1
Longest subarray with sum K	https://www.geeksforgeeks.org/problems/longest-sub-array-with-sum-k0809/1
Count of subarray with sum K	https://leetcode.com/problems/subarray-sum-equals-k/
Count of subarrays with Sum divisible by K	https://leetcode.com/problems/subarray-sums-divisible-by-k/
Longest subarray with equal number of 0s and 1s	https://leetcode.com/problems/contiguous-array/description/
Count of subarray with equal number of 0s and 1s	https://www.geeksforgeeks.org/problems/count-subarrays-with-equal-number-of-1s-and-0s-1587115620/1
LRU cache	https://leetcode.com/problems/lru-cache/description
LFU cache	https://leetcode.com/problems/lfu-cache/description/
Maximum Frequency stack	https://leetcode.com/problems/maximum-frequency-stack/description/
Insert delete getRandom in O(1)	https://leetcode.com/problems/insert-delete-getrandom-o1/description/

MORE QUESTION ON HASHMAP HEAP WILL BE DONE DURING ARRAYS AND STRINGS

Heap/PriorityQueue

	Link
Kth Largest element	https://leetcode.com/problems/kth-largest-element-in-an-array/
Kth Smallest element	https://www.geeksforgeeks.org/problems/kth-smallest-element5635/1
Sort nearly sorted (K-sorted) array	https://www.geeksforgeeks.org/problems/nearly-sorted-1587115620/1
Median of data stream	https://leetcode.com/problems/find-median-from-data-stream/description/
Merge k sorted linked-lists	https://leetcode.com/problems/merge-k-sorted-lists/
Merge k sorted arrays	https://www.geeksforgeeks.org/problems/merge-k-sorted-arrays/1

TREES

	Link
Generic Trees	
Construction of generic tree	
Display of generic tree	
Size of Tree	
Maximum of Tree	
Height of Tree	
Euler traversal of tree	
Level order traversal (3 methods)	
Zig zag traversal	
Mirror a generic Tree	https://www.geeksforgeeks.org/problems/mirror-of-n-ary-tree/
Remove leaf nodes	https://www.geeksforgeeks.org/remove-all-leaf-nodes-from-a-generic-tree-or-n-ary-tree/
Linearise a GT, O(N2)	Similar to linked_list: https://leetcode.com/problems/flatten-a-multilevel-doubly-linked-list/description/
Linearise a GT, O(N)	
Find in Generic Tree	
Node to root Path	
Lowest common Ancestor (LCA) in generic tree	
Is Tree Mirror	
Is Tree Symmetric	https://www.geeksforgeeks.org/problems/check-if-given-generic-n-ary-tree-is-symmetric-horizontally/
Binary Trees	
Construct a binary Tree	
Display a binary Tree	
Size, sum, max, height of tree	
Depth of tree	https://leetcode.com/problems/maximum-depth-of-binary-tree/
Sum of BT	https://www.geeksforgeeks.org/problems/sum-of-binary-tree/1
Size of Binary Tree	https://www.geeksforgeeks.org/problems/size-of-binary-tree/1
Preorder Traversal of BT using recursion	https://leetcode.com/problems/binary-tree-preorder-traversal/description/
Postorder Traversal using recursion	https://leetcode.com/problems/binary-tree-postorder-traversal/description/
Inorder Traversal using recursion	https://leetcode.com/problems/binary-tree-inorder-traversal/description/
Preorder Traversal of BT using stack	https://leetcode.com/problems/binary-tree-preorder-traversal/description/
Postorder Traversal using stack	https://leetcode.com/problems/binary-tree-postorder-traversal/description/
Inorder Traversal using stack	https://leetcode.com/problems/binary-tree-inorder-traversal/description/
Node to root path in Binary Tree	
Get K levels down in Binary Tree	
Get Nodes at K distance	https://leetcode.com/problems/all-nodes-distance-k-in-binary-tree/
LCA of binary tree	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree/
LCA in O(1) Space	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree/
Minimum time required to burn tree if one node is on fire	https://www.geeksforgeeks.org/problems/burning-tree/1
Minimum time required to burn tree with water nodes	Followup of https://www.geeksforgeeks.org/problems/burning-tree/1
Get nodes at K distance on O(1 + no) space	https://leetcode.com/problems/all-nodes-distance-k-in-binary-tree/
Delete leaf nodes with a given value	https://www.geeksforgeeks.org/delete-leaves-with-a-given-value/problem
Construct tree from preorder and inorder	https://leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal/description/
Construct tree from postorder and inorder	https://leetcode.com/problems/construct-binary-tree-from-inorder-and-postorder-traversal/description/
Construct tree from preorder and postorder	https://leetcode.com/problems/construct-binary-tree-from-preorder-and-postorder-traversal/
Maximum width of Binary Tree	https://leetcode.com/problems/maximum-width-of-binary-tree/description/
Left View of Binary Tree	https://www.geeksforgeeks.org/problems/left-view-of-binary-tree/1
Right view of Binary Tree	https://leetcode.com/problems/binary-tree-right-side-view/description/
Vertical order of Binary Tree without preference	https://www.geeksforgeeks.org/problems/print-a-binary-tree-in-vertical-order/1
Vertical order of BT with Preference	https://leetcode.com/problems/vertical-order-traversal-of-a-binary-tree/description/
Vertical Sum of Binary Tree	http://geeksforgeeks.org/problems/vertical-sum/1
Bottom View of Binary Tree	https://www.geeksforgeeks.org/problems/bottom-view-of-binary-tree/1
Sum of Bottom View of Binary Tree	https://www.geeksforgeeks.org/sum-of-nodes-in-bottom-view-of-binary-tree/
Top View of Binary Tree	https://www.geeksforgeeks.org/problems/top-view-of-binary-tree/1
Sum of nodes of top view of binary Tree	https://www.geeksforgeeks.org/sum-of-nodes-in-top-view-of-binary-tree/
Diagnoal View of Binary Tree	https://www.geeksforgeeks.org/problems/diagonal-traversal-of-binary-tree/1
Diameter of Binary Tree (All 3 Methods)	https://leetcode.com/problems/diameter-of-binary-tree/description/
Max path sum from leaf to leaf	https://www.geeksforgeeks.org/problems/max-path-sum/1
Max path sum from any node to any node	https://leetcode.com/problems/binary-tree-maximum-path-sum/description/
Morris Traversal Implementation	
Morris Inorder Traversal	https://leetcode.com/problems/binary-tree-inorder-traversal/description/
Morris Preorder Traversal	https://leetcode.com/problems/binary-tree-preorder-traversal/description/
Morris Postorder Traversal	https://leetcode.com/problems/binary-tree-postorder-traversal/description/
HasPath sum	https://leetcode.com/problems/path-sum/
get all paths with target sum	https://leetcode.com/problems/path-sum-ii/
Binary tree to doubly LL	https://www.geeksforgeeks.org/problems/binary-tree-to-dll/1, https://leetcode.com/problems/convert-binary-search-tree-to-sorted-doubly-linked-list/description/
Flatten Binary Tree to LinkedList - copy of flatten generic tree	https://leetcode.com/problems/flatten-binary-tree-to-linked-list/description/
Flatten Binary Tree to LinkedList - move in reverse preoder recursively	https://leetcode.com/problems/flatten-binary-tree-to-linked-list/description/
Populate next right pointer of each node in a perfect binary tree	https://leetcode.com/problems/populating-next-right-pointers-in-each-node/description/
Populate next right pointer of each node in non-perfect binary tree	https://leetcode.com/problems/populating-next-right-pointers-in-each-node-ii/description/
Serialize and deserialize Binary Tree	https://leetcode.com/problems/serialize-and-deserialize-binary-tree/description/
Distribute coins in binary tree	https://leetcode.com/problems/distribute-coins-in-binary-tree/description/
Binary tree cameras	https://leetcode.com/problems/binary-tree-cameras/
Binary tree coloring game (Greedy)	https://leetcode.com/problems/binary-tree-coloring-game/description/
Binary Tree Pruning	https://leetcode.com/problems/binary-tree-pruning/description/
Height of Binary Tree After Subtree Removal Queries	https://leetcode.com/problems/height-of-binary-tree-after-subtree-removal-queries/description/
Binary Search Trees	
Construct BST	
Display a binary Tree	
Size, sum	
Max Of BST	
Min of BST	
Search BST	https://leetcode.com/problems/search-in-a-binary-search-tree/description/
LCA of BST	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/description/
Insert node in BST	https://leetcode.com/problems/insert-into-a-binary-search-tree/description/
Delete Node in BST	https://leetcode.com/problems/delete-node-in-a-bst/
Trim BST on the basis of lower limit, upper limit	https://leetcode.com/problems/trim-a-binary-search-tree/description/
Convert bst to greater Tree	https://leetcode.com/problems/convert-bst-to-greater-tree/description/
Find 2 sum pair in BST	https://leetcode.com/problems/two-sum-iv-input-is-a-bst/description/
Find 2 sum pair in BST iteratively	https://leetcode.com/problems/two-sum-iv-input-is-a-bst/description/
Construct BST from preorder	https://leetcode.com/problems/construct-binary-search-tree-from-preorder-traversal/description/
Construct BST from postorder	https://www.geeksforgeeks.org/problems/construct-bst-from-post-order/1
Validate BST using BST Pair	https://leetcode.com/problems/validate-binary-search-tree/description/
Maximum Sum BST in Binary Tree	https://leetcode.com/problems/maximum-sum-bst-in-binary-tree/description/
Validate BST Using Inorder traversal	https://leetcode.com/problems/validate-binary-search-tree/description/
Validate BST Using inorder traversal but O(1) Space	https://leetcode.com/problems/validate-binary-search-tree/description/
Recover BST	https://leetcode.com/problems/recover-binary-search-tree/description/
BST Iterator	https://leetcode.com/problems/binary-search-tree-iterator/
Kth Smallest element in BST	https://leetcode.com/problems/kth-smallest-element-in-a-bst/
Predecessor and Successor of BST key using recursion	https://www.geeksforgeeks.org/problems/predecessor-and-successor/1
Predecessor and Successor of BST key using iteration	https://www.geeksforgeeks.org/problems/predecessor-and-successor/1
AVL Trees / Red-Black Trees	
Check if tree is balanced	https://leetcode.com/problems/balanced-binary-tree/description/
AVL Tree implementation	
Balance a BST	https://leetcode.com/problems/balance-a-binary-search-tree/
Insert into AVL	https://www.geeksforgeeks.org/problems/avl-tree-insertion/1

Graph

	Link
What, why Graph	
Construction of graph - Adjacency matrix, list	
Check if there is a path between given source and destination	
Print all paths between given source and destination	
Find path with minimum weight between source and destination	
Find path with maximum weight between source and destination	
Find kth largest path weight between source and destination	
Get all the connected components	https://www.geeksforgeeks.org/problems/connected-components-in-an-undirected-graph/1
Check if the whole graph is connected	
Print all hamiltonian Paths and Cycle	
Check if there is a hamiltonian path	https://www.geeksforgeeks.org/problems/hamiltonian-path2522/1
DFS Problems	
Number of islands	https://leetcode.com/problems/number-of-islands/description/
Max Area of island	https://leetcode.com/problems/max-area-of-island/
Surrounded regions	https://leetcode.com/problems/surrounded-regions/description/
Perimeter of island	https://leetcode.com/problems/island-perimeter/description/
Number of provinces	https://leetcode.com/problems/number-of-provinces/
Count sub islands	https://leetcode.com/problems/count-sub-islands/description/
Number of distinct islands	https://www.geeksforgeeks.org/problems/number-of-distinct-islands/1
Implementation of Topological Sort For DAG	
Implementation of Topological Sort For Generic Directed Graph	
Is course scheduling possible	https://leetcode.com/problems/course-schedule/description/
Find course Schedule	https://leetcode.com/problems/course-schedule-ii/description/
Find eventual safe states	https://leetcode.com/problems/find-eventual-safe-states/description/
Implement Kosaraju Algorithm	https://www.geeksforgeeks.org/problems/strongly-connected-components-kosarajus-algo/1
Detonate the maximum bombs	https://leetcode.com/problems/detonate-the-maximum-bombs/description/
M coloring problem	https://www.geeksforgeeks.org/problems/m-coloring-problem-1587115620/1
Breadth First Search (BFS)	
BFS Implementation	
BFS Implementation with cycle detection	
As Far from land as possible	https://leetcode.com/problems/as-far-from-land-as-possible/description/
Zero-One Matrix	01 Matrix - LeetCode
Rotting oranges	https://leetcode.com/problems/rotting-oranges/
Shortest Path in binary Matrix	https://leetcode.com/problems/shortest-path-in-binary-matrix/description/
Walls and Gates	https://www.lintcode.com/problem/663/
Shortest bridge	https://leetcode.com/problems/shortest-bridge/description/
Kahn's Algorithm Implementation	
Is course scheduling possible using BFS	https://leetcode.com/problems/course-schedule/description/

	Find course Schedule using BFS	https://leetcode.com/problems/course-schedule/description/
	Longest Increasing Path	https://leetcode.com/problems/longest-increasing-path-in-a-matrix/
	What is Bipartite Graph	
	Is Graph bipartite	https://leetcode.com/problems/is-graph-bipartite/
	Possible bipartition	https://leetcode.com/problems/possible-bipartition/
	Word ladder without graph	https://leetcode.com/problems/word-ladder/
	Word ladder with graph	https://leetcode.com/problems/word-ladder/
	Word ladder 2	https://leetcode.com/problems/word-ladder-ii/description/
	Bus Routes	https://leetcode.com/problems/bus-routes/description/

Union Find/Disjoint-set-union (DSU)

	DSU implementation, path compression, union by rank	
	Redundant Connections	https://leetcode.com/problems/redundant-connection/
	Lexicographically smallest equivalent string	https://leetcode.com/problems/lexicographically-smallest-equivalent-string/
	Number of provinces using DSU	https://leetcode.com/problems/number-of-provinces/
	Max area of island using DSU	https://leetcode.com/problems/max-area-of-island/
	Number of islands 2	https://leetcode.ca/all/305.html
	Similar string groups with the help of array indices	https://leetcode.com/problems/similar-string-groups/description/
	Similar string groups without the help of array indices	https://leetcode.com/problems/similar-string-groups/description/
	Satisfiability of Equality Equations	https://leetcode.com/problems/satisfiability-of-equality-equations/
	Count servres that communicate	https://leetcode.com/problems/count-servers-that-communicate/description/
	Regions cut by slashes	https://leetcode.com/problems/regions-cut-by-slashes/description/
	Number of operations to make network connected	https://leetcode.com/problems/number-of-operations-to-make-network-connected
	Smallest string with swaps	https://leetcode.com/problems/smallest-string-with-swaps/description/
	Accounts merge	https://leetcode.com/problems/accounts-merge/
	Making a large island	https://leetcode.com/problems/making-a-large-island/description/
	Minimize malware spread	https://leetcode.com/problems/minimize-malware-spread/description/

Minimum Spanning Tree

	What is spanning Tree, what is minimum spanning Tree	
	Create MST using DSU (Kruskal)	

Graph Algorithms

	Kruskal Algorithm Implementation	
	Min cost to connect all points	https://leetcode.com/problems/min-cost-to-connect-all-points/description/
	Optimize Water distribution in a village	https://leetcode.ca/all/1168.html
	Connecting cities with minimum cost	https://leetcode.ca/all/1135.html
	Minimize maximum component cost	https://leetcode.com/problems/minimize-maximum-component-cost/description/
	Build MST using Prime Algorithm	
	Find cost of MST	https://www.geeksforgeeks.org/problems/minimum-spanning-tree/1
	Dijkstra Algorithm Implementation	
	Network delay time	https://leetcode.com/problems/network-delay-time/
	Path with minimum effort	https://leetcode.com/problems/path-with-minimum-effort/
	Minimum cost to reach destination in time	https://leetcode.com/problems/minimum-cost-to-reach-destination-in-time/
	Number of ways to arrive at destination	https://leetcode.com/problems/number-of-ways-to-arrive-at-destination/description/
	Lexicographically shortest maze path to hole	https://leetcode.ca/all/499.html
	Swim in rising water	https://leetcode.com/problems/swim-in-rising-water/description/
	Network recovery Pathways	Network Recovery Pathways - LeetCode — Leetcode Weekly Contest 360	Network Recovery Pathways Detailed Solution with Dry run
	Bellman ford Implementation	
	Cheapest flights within k stops	https://leetcode.com/problems/cheapest-flights-within-k-stops/
	Floyd Warshall Algorithm	https://www.geeksforgeeks.org/problems/implementing-floyd-warshall/1
	Implement floyd warshall	https://www.geeksforgeeks.org/problems/implementing-floyd-warshall/0/1062421
	Articulation point(Tarjan's Algorithm (optional)	https://www.geeksforgeeks.org/problems/articulation-points-or-cut-vertices-in-a-graph/1
	Mother Vertex	https://www.geeksforgeeks.org/problems/mother-vertex/1

Mixed of DFS/BFS/DSU

	Pacific Atlantic water flow	https://leetcode.com/problems/pacific-atlantic-water-flow/description/
	Clone graph	https://leetcode.com/problems/clone-graph/description/
	Last Day where you can still cross	https://leetcode.com/problems/last-day-where-you-can-still-cross/description
	Shortest path with alternating colors	https://leetcode.com/problems/shortest-path-with-alternating-colors/description/

Dynamic Programming

	What is dynamic Programming	Dynamic Programming || How to solve any DP problem || DP series Part 1
	How Memoization works	Dynamic Programming Tricks || Leetcode 377 Combination Sum # || Coin Change || Hack to s
	Steps to solve any DP problem	Dynamic Programming Tricks || Leetcode 377 Combination Sum # || Coin Change || Hack to s
	Find nth Term of fibonacci series	Dynamic Programming || How to solve any DP problem || DP series Part 1
	Friends pairing using dp	https://www.geeksforgeeks.org/problems/friends-pairing-problem/0/0/1
	Climbing stairs	https://leetcode.com/problems/climbing-stairs/
	Unique maze path	https://leetcode.com/problems/unique-paths/description/
	Unique maze path with diagonal moves	
	Min maze path sum	https://leetcode.com/problems/minimum-path-sum/description/
	GoldMine	Gold Mine Problem	Practice	GeeksforGeeks
	Unique paths 2	https://leetcode.com/problems/unique-paths-ii/
	Min cost climbing stairs	https://leetcode.com/problems/min-cost-climbing-stairs/description/
	Min cost climbing stairs in O(1) space	https://leetcode.com/problems/min-cost-climbing-stairs/description/
	Decode ways	https://leetcode.com/problems/decode-ways/description/
	Decode ways in O(1) space	https://leetcode.com/problems/decode-ways/description/
	Decode ways 2	https://leetcode.com/problems/decode-ways-ii/
	Decode ways 2 in O(1) space	https://leetcode.com/problems/decode-ways-ii/
	Jump game 2	https://leetcode.com/problems/jump-game-ii/description/

Target Sum subset

	Coin Change Permutation using yes/no call	
	Coin Change Permutation using for loop	
	Coin change combination using 2D dp	https://leetcode.com/problems/coin-change-ii/description/
	Coin change combination using 1D dp	https://leetcode.com/problems/coin-change-ii/description/
	Combination sum 4	https://leetcode.com/problems/combination-sum-iv/
	Minimum coins required	Coin Change - LeetCode
	Find if any subset with given target exists	https://www.geeksforgeeks.org/problems/subset-sum-problem-2e294883df/1
	Count of subsets with sum = target	https://www.geeksforgeeks.org/problems/perfect-sum-problem/0/1
	Is it possible to partition array into two subsets of equal sum	https://leetcode.com/problems/partition-equal-subset-sum/
	0-1 Knapsack	https://www.geeksforgeeks.org/problems/0-1-knapsack-problem/0/1
	Find a string with items picked to get the maximum profit in knapsack	https://www.geeksforgeeks.org/problems/0-1-knapsack-problem/0/1
	Unbounded knapsack	Knapsack with Duplicate Items	Practice	GeeksforGeeks
	Find a string with items picked to get the maximum profit in unbounded knapsack	Knapsack with Duplicate Items	Practice	GeeksforGeeks
	Target Sum with positive negative values	Target Sum - LeetCode

DP on Strings

	Longest palindromic substring	https://leetcode.com/problems/longest-palindromic-substring/
	Number of palindromic substring	https://leetcode.com/problems/palindromic-substrings/description/
	Longest palindromic subsequence using Recursion	Longest Palindromic Subsequence - LeetCode
	Longest palindromic subsequence using memo	Longest Palindromic Subsequence - LeetCode
	Longest palindromic subsequence using tabulation	Longest Palindromic Subsequence - LeetCode
	Generate longest palindromic subsequence using tabulation	Longest Palindromic Subsequence - LeetCode
	Distinct subsequences	Distinct Subsequences - LeetCode
	Longest common subsequence length	https://leetcode.com/problems/longest-common-subsequence/description/
	Longest common subsequence string using tabulation	Longest Common Subsequence - LeetCode
	Delete operation for 2 strings	https://leetcode.com/problems/delete-operation-for-two-strings/
	Delete operation but with ASCII values	Minimum ASCII Delete Sum for Two Strings - LeetCode
	Maximum uncrossed lines	https://leetcode.com/problems/uncrossed-lines/description/
	Edit distance	Edit Distance - LeetCode
	WildCard matching	Wildcard Matching - LeetCode
	Regular expression Matching	Regular Expression Matching - LeetCode
	Distinct Subsequences 2	Distinct Subsequences II - LeetCode
	Count sub-sHype a=b-ck	Count subsequences of type a*i, b*j, c*k	Practice	GeeksforGeeks

Longest Increasing Subsequence(LIS) set

	Longest increasing subsequence recursion + memoization	https://leetcode.com/problems/longest-increasing-subsequence/description/
	Longest increasing subsequence tabulation	https://leetcode.com/problems/longest-increasing-subsequence/description/
	Longest Decreasing subsequence with dp[i] = LDS ending with num[i]	https://www.geeksforgeeks.org/problems/longest-decreasing-subsequence/1
	Longest Decreasing subsequence with dp[i] = LDS starting with num[i]	https://www.geeksforgeeks.org/problems/longest-decreasing-subsequence/1
	Longest Bitonic subsequence	https://www.geeksforgeeks.org/problems/longest-bitonic-subsequence0824/1
	Maximum sum increasing subsequence	https://www.geeksforgeeks.org/problems/maximum-sum-increasing-subsequence4749/1
	Maximum sum bitonic subsequence	Maximum Sum Bitonic Subsequence	Practice	GeeksforGeeks
	Minimum deletion to make array sorted	https://www.geeksforgeeks.org/problems/minimum-number-of-deletions-to-make-a-sorted-sequence5824/1
	Minimum number of removal to make bitonic array	Minimum Number of Removals to Make Mountain Array - LeetCode
	Number of LIS	http://leetcode.com/problems/number-of-longest-increasing-subsequence/description/
	Longest Arithmetic subsequence	https://leetcode.com/problems/longest-arithmetic-subsequence/description/
	Longest increasing subsequence in NLogN	https://leetcode.com/problems/longest-increasing-subsequence/description/
	Russian doll envelopes	https://leetcode.com/problems/russian-doll-envelopes/
	Maximum height buy stacking cuboids	https://leetcode.com/problems/maximum-height-by-stacking-cuboids/description/
	Maximum length of pair chain	Maximum Length of Pair Chain - LeetCode
	Maximum profit in job scheduling	Maximum Profit in Job Scheduling - LeetCode

Cut Type DP set

	Matrix chain multiplication	Matrix Chain Multiplication	Practice	GeeksforGeeks
	Brackets in MCM	https://www.geeksforgeeks.org/problems/brackets-in-matrix-chain-multiplication/1/0/41
	Burst Balloons	https://leetcode.com/problems/burst-balloons/description/
	Minimum score triangulation	https://leetcode.com/problems/minimum-score-triangulation-of-polygon/description/
	Minimum cost to cut a stick	https://leetcode.com/problems/minimum-cost-to-cut-a-stick/description/
	Palindrome Partitioning	https://leetcode.com/problems/palindrome-partitioning-iii/description/
	Palindrome Partitioning in 2 parts(Optional if you have time)	https://leetcode.com/problems/palindrome-partitioning-iv
	Partition Array for maximum sum	https://leetcode.com/problems/partition-array-for-maximum-sum/description/
	Optimal BST	https://www.geeksforgeeks.org/problems/optimal-binary-search-tree2214/1

Buy and Sell Stocks

	Buy and Sell Stocks in O(N) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/
	Buy and Sell Stocks in O(1) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/
	Buy and Sell Stocks 2 in O(N) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-ii/description/
	Buy and Sell Stocks 2 in O(1) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-ii/description/
	Buy and Sell Stocks with Transaction fees	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-with-transaction-fee/description/
	Buy and Sell Stocks with Cooldown	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-with-cooldown/description/
	Buy and Sell Stocks 3 in O(N^2) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-iii/
	Buy and Sell Stocks 3 in O(1) space	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-iii/
	Buy and Sell Stocks 4	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-iv/description/

Miscellaneous Set

	House robber	https://leetcode.com/problems/house-robber/
	House robber 2	https://leetcode.com/problems/house-robber-ii/description/
	Count square submatrices	https://leetcode.com/problems/count-square-submatrices-with-all-ones/description/
	Maximum area of square submatrice with all ones	https://leetcode.com/problems/maximal-square/description/
	Count submatrices with all Ones	https://leetcode.com/problems/count-submatrices-with-all-ones
	Cherry Pickup 2	https://leetcode.com/problems/cherry-pickup/description
	Cherry Pickup 2 in n*m*m space	https://leetcode.com/problems/cherry-pickup/description
	Cherry Pickup 2 in 2*n*m space	https://leetcode.com/problems/cherry-pickup/description
	Cherry Pickup	https://leetcode.com/problems/cherry-pickup
	Stone game	https://leetcode.com/problems/stone-game/description/
	Stone game 4	https://leetcode.com/problems/stone-game-iv/description/
	Ones and Zeroes	https://leetcode.com/problems/ones-and-zeroes/description/
	Paint N houses	https://www.geeksforgeeks.org/problems/painting-walls/0/#1704451

Arrays & Strings

Problems based on searching, sorting, sliding/window etc

	Segregate positive and negative number without maintaining order	
	Segregate 0s and 1s	Segregate 0s and 1s	Practice	GeeksforGeeks
	Segregate 0s , 1s and 2s (Sort colors)	Sort Colors - LeetCode

Sliding Window

	Longest substring without repeating characters	Longest Substring Without Repeating Characters - LeetCode
	Longest substring with atmost 2 distinct characters	Longest Substring with At Most Two Distinct Characters - LeetCode
	Longest Substring with exact K distinct	Longest Substring with K Unique Characters	Practice	GeeksforGeeks
	Maximum number of vowels in a substring of K length	Maximum Number of Vowels in a Substring of Given Length - LeetCode
	Sliding Window maximum (Already covered twice in Stack and then Queue, use it)	Sliding Window Maximum - LeetCode
	Fruit into basket	Fruit Into Baskets - LeetCode
	Largest subarray with zero sum	Largest subarray with 0 sum	Practice	GeeksforGeeks
	Count of subarray with zero sum	Zero Sum Subarrays	Practice	GeeksforGeeks
	Longest subarray with sum K	Longest Subarray with Sum K	Practice	GeeksforGeeks
	Count of subarray with sum K	Subarray Sum Equals K - LeetCode
	Count of subarrays with Sum divisible by K	Subarray Sums Divisible by K - LeetCode
	Count of subarrays with XOR K	Count Subarrays with given XOR	Practice	GeeksforGeeks
	Count of subarray with equal number of 0s and 1s	Contiguous Array - LeetCode
	Count of subarray with equal number of 0s and 1s	Subarrays with equal 1s and 0s	Practice	GeeksforGeeks
	Longest consecutive subsequence	Longest Consecutive Sequence - LeetCode

Searching and Sorting

	Binary Search	Binary Search - LeetCode
	Find Insertion position	Search Insert Position - LeetCode
	First and last Index of elements in sorted array	Find First and Last Position of Element in Sorted Array - LeetCode
	Find K closest elements to target	Find K Closest Elements - LeetCode
	4th Missing Positive Number	4th Missing Positive Number - LeetCode
	Find minimum in rotated sorted array	Find Minimum in Rotated Sorted Array - LeetCode
	Find minimum in rotated sorted array with duplicate elements	Find Minimum in Rotated Sorted Array II - LeetCode
	Search in rotated sorted array	Search in Rotated Sorted Array - LeetCode
	Search in rotated sorted array with duplicate elements	Search in Rotated Sorted Array II - LeetCode
	Median of two Sorted Arrays	Median of Two Sorted Arrays - LeetCode
	Aggressive cows	Aggressive Cows	Practice	GeeksforGeeks
	Koko Eating Bananas	Koko Eating Bananas - LeetCode
	Maximum candies allocated to K children	Maximum Candies Allocated to K Children - LeetCode
	Minimum number of days to make m bouquets	Minimum Number of Days to Make m Bouquets - LeetCode
	Capacity to ship packages with D days	Capacity To Ship Packages Within D Days
	Minimum time to repair cars	Minimum Time to Repair Cars - LeetCode
	Allocate minimum number of pages	Allocate Minimum Pages	Practice	GeeksforGeeks
		https://www.geeksforgeeks.org/problems/minimize-max-distance-to-gas-station/1
	Split array largest sum	https://leetcode.com/problems/split-array-largest-sum/description/

Sorting

	Bubble sort Algorithm	Bubble Sort Algorithm - GeeksforGeeks
	Insertion sort Algorithm	Insertion Sort Algorithm - GeeksforGeeks
	Merge two sorted arrays	
	Merge sort	Merge Sort	Practice	GeeksforGeeks
	Inversion count	Count Inversions	Practice	GeeksforGeeks
	Partition array according to pivot	
	Quick Sort	Quick Sort	Practice	GeeksforGeeks

		Miscellaneous Problems	
	☐	Two Sum	Two Sum - LeetCode
	☐	Two Sum 2 in O(1) Space	Two Sum II - Input Array is Sorted - LeetCode
	☐	All pairs for Two Sum	All distinct pairs with given sum | Practice | GeeksforGeeks
	☐	3 Sum	3Sum - LeetCode
	☐	Four sum	4Sum - LeetCode
	☐	Ksum (k= 5 sum, 6 sum ...)	KSum - LeetCode
	☐	Count pairs with given sum	Count pairs with given sum | Practice | GeeksforGeeks
	☐	Count quadruplets with given sum	4Sum II - LeetCode
	☐	Majority Element in O(N) time and O(N) space	Majority Element - LeetCode
	☐	Majority Element in O(N) time and O(1) space	Majority Element - LeetCode
	☐	Majority Element 2	Majority Element II - LeetCode
	☐	Next permutation	Next Permutation - LeetCode
	☐	Merge intervals without using Stack	Merge Intervals - LeetCode
	☐	Missing and Repeating Number in O(1) Space and O(N) time	Missing And Repeating | Practice | GeeksforGeeks
	☐	Maximum product subarray	Maximum Product Subarray - LeetCode
	☐	Reverse words in a string	Reverse Words in a String - LeetCode
	☐	Gas station	Gas Station - LeetCode